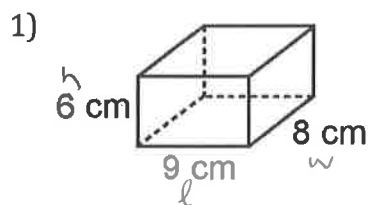
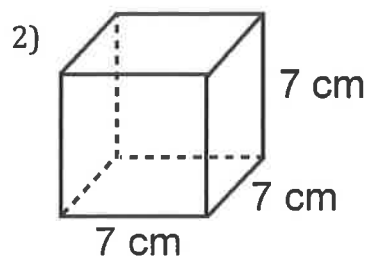


8.3 Surface Area of Prisms and Pyramids

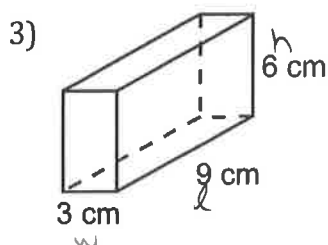
Part 1: Find the Surface Area of the Following Objects:



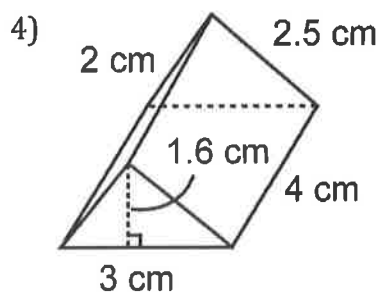
$$\begin{aligned} SA &= 2lw + 2wh + 2lh \\ &= 2(9)(8) + 2(8)(6) + 2(9)(6) \\ &= 144 + 96 + 108 \\ &= 348 \text{ cm}^2 \end{aligned}$$



$$\begin{aligned} SA &= 6s^2 \\ &= 6(7)^2 \\ &= 294 \text{ cm}^2 \end{aligned}$$

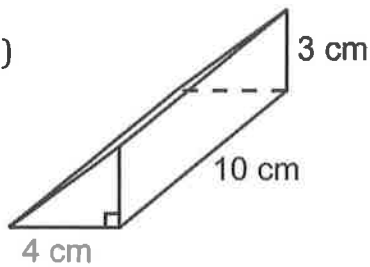


$$\begin{aligned} SA &= 2lw + 2wh + 2lh \\ &= 2(9)(3) + 2(3)(6) + 2(9)(6) \\ &= 54 + 36 + 108 \\ &= 198 \text{ cm}^2 \end{aligned}$$



$$\begin{aligned} SA &= 2\left(\frac{1}{2}\right)(3)(4) + (4)(2.5) + (4)(3) + (4)(2) \\ &= 4.8 + 10 + 12 + 8 \\ &= 34.8 \text{ cm}^2 \end{aligned}$$

5)



$$c^2 = a^2 + b^2$$

$$c^2 = 4^2 + 3^2$$

$$c^2 = 16 + 9$$

$$c^2 = 25$$

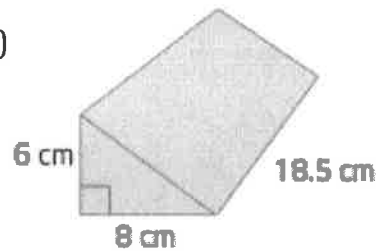
$$c = 5$$

$$SA = 2\left(\frac{1}{2}\right)(4)(3) + (4)(10) + (3)(10) + (5)(10)$$

$$= 12 + 40 + 30 + 50$$

$$= 132 \text{ cm}^2$$

6)



$$c^2 = a^2 + b^2$$

$$c^2 = 6^2 + 8^2$$

$$c^2 = 100$$

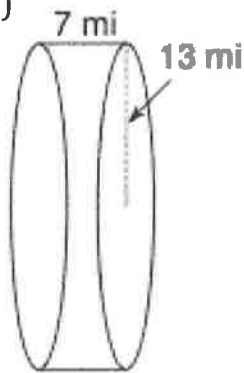
$$c = 10$$

$$SA = 2\left(\frac{1}{2}\right)(8)(6) + (8.5)(10) + (6)(10) + (10)(18.5)$$

$$= 48 + 85 + 60 + 185$$

$$= 492 \text{ cm}^2$$

7)



$$SA = 2\pi r^2 + 2\pi rh$$

$$= 2\pi(13)^2 + 2\pi(13)(7)$$

$$= 1061.9 + 571.8$$

$$= 1633.7 \text{ mi}^2$$

8) A cylinder with a diameter of 10cm
and a height of 25 cm

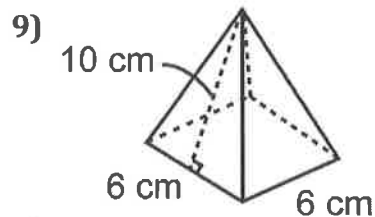
$$r = 5$$

$$SA = 2\pi r^2 + 2\pi rh$$

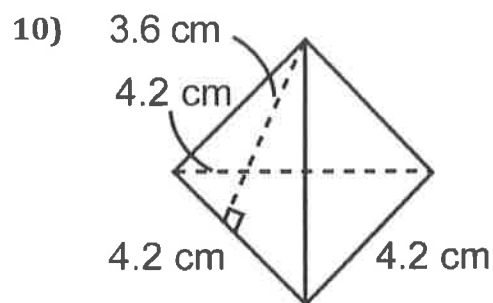
$$= 2\pi(5)^2 + 2\pi(5)(25)$$

$$= 157.1 + 785.4$$

$$= 942.5 \text{ cm}^2$$



$$\begin{aligned}
 SA &= (6)^2 + 4\left(\frac{1}{2}\right)(6)(10) \\
 &= 36 + 120 \\
 &= 156 \text{ cm}^2
 \end{aligned}$$



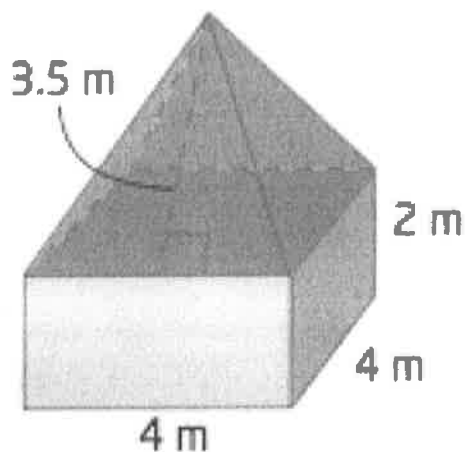
$$\begin{aligned}
 SA &= 4\left(\frac{1}{2}\right)(4.2)(3.6) \\
 &= 30.24 \text{ cm}^2
 \end{aligned}$$

Part 2: Applications

11) Find the side length of a cube with a surface area of 726 cm^2

$$\begin{aligned}
 SA &= 6s^2 \\
 726 &= 6s^2 \\
 121 &= s^2 \\
 s &= \sqrt{121} \\
 s &= 11 \text{ cm}
 \end{aligned}$$

12) Adam has built a garden shed in the shape shown:



- a) Adam plans to paint the outside of the shed, including the roof but not the floor. One can of paint covers 4 m^2 . How many cans of paint will Adam need?

$$\begin{aligned} SA &= 4(4)(2) + 4\left(\frac{1}{2}\right)(4)(3.5) \\ &= 32 + 28 \\ &= 60 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Cans of Paint} &= 60 \div 4 \\ &= 15 \end{aligned}$$

He will need 15 cans of paint.

- b) If one can costs \$16.95, what will the cost be including the 15% tax?

$$\begin{aligned} \text{Cost} &= 1.15(16.95 \times 15) \\ &= \$292.39. \end{aligned}$$